Project Interim Report

On

Design and Implementation of a Digital Stopwatch using Object Oriented Programming

RU-100-01-00012

Object Oriented Programming with Java

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ABSTRACT

- This project is based on the design and implementation of a Digital Stopwatch using Core Java and Object-Oriented Programming (OOP) concepts. A stopwatch is a time-measuring device used to record the duration of events with precision.
- The need for this project arises from the requirement of accurate and efficient time tracking in various real-life applications such as sports, laboratories, and productivity monitoring. Manual timing methods are often prone to errors, while digital systems provide better accuracy and usability.
- The main objective of this project is to simulate a real stopwatch system that can start, stop, reset, and display elapsed time. The project also demonstrates important OOP concepts like classes, objects, encapsulation, and methods. This system provides a simple and user-friendly interface for measuring time intervals and can be further enhanced with graphical user interface and advanced features.

INTRODUCTION

- In today's digital world, time management plays a very important role in almost every field. A stopwatch is a basic yet powerful tool used to measure the duration of events. This project focuses on developing a Digital Stopwatch using Java programming language.
- The system is designed using Object-Oriented Programming concepts, which help in organizing the program into classes and objects. This makes the code reusable, easy to understand, and maintainable.
- The stopwatch performs basic operations such as starting the timer, stopping it, resetting it, and displaying the current elapsed time. The system continuously updates time using logical conditions and looping structures.

❖ Objectives of the Project

1. To design a digital stopwatch using Java
2. To implement Object-Oriented Programming concepts
3. To provide accurate time measurement
4. To reduce human errors in manual timing

scope

The scope of this project is limited to basic stopwatch functionalities, but it has wide applications and future possibilities.

This system can be used in:

- Sports events for tracking time
- Educational institutions for experiments
- Daily activities for time management

Future Scope

- Integration with GUI (Graphical User Interface)
- Mobile application development
- Web-based stopwatch system
- Database storage for saving records
- Adding lap time and split time features

SYSTEM REQUIRIMENTS

❖ **Hardware Requirements**

1) Computer/Laptop
2) Minimum 4GB RAM
3) Keyboard and Monitor

➔ **Software Requirements**

Java Development Kit (JDK)

IDE (Eclipse / VS Code / IntelliJ IDEA)

Operating System (Windows/Linux/Mac)

METHODOLOGY

- The methodology describes the step-by-step working process of the system.

1. The program starts execution
2. A menu is displayed to the user
3. User selects an option
4. Based on input, corresponding function is executed
5. Stopwatch starts counting time using loop
6. Time is updated every second
7. User can stop or reset the timer
8. Program continues until exit option is selected

- This approach ensures smooth functioning and user interaction.

FLOWCHART

★ Flowchart Steps

➢ Start
➢ Display Menu
➢ Accept User Input
➢ Perform Operation (Start/Stop/Reset)
➢ Loop until Exit
➢ End

➢ Class Diagram

➢ Class Name: Stopwatch

➔ Data Members:

- int hours
- int minutes
- int seconds

❖ Member Functions:

- start() → starts timer
- stop() → stops timer
- reset() → resets timer
- displayTime() → displays time

★ Main Class:

- Contains main method
- Handles user input
- Calls Stopwatch methods

IMPLEMENTATION

The project is implemented using Java programming language with proper use of classes and objects.

1. Stopwatch Class

This class contains all the variables and methods required for time calculation. It stores time values and updates them accordingly.

2. Main Class

This class controls the execution of the program. It displays menu options and takes input from the user.

Working Logic

- A loop runs continuously
- Time increments every second
- When seconds reach 60 → minutes increase
- When minutes reach 60 → hours increase
- Display function prints formatted time

INPUT/OUTPUT

----- MENU -----

1. Start
2. Stop
3. Reset
4. Exit

Enter your choice: 1

Stopwatch Started...

Time: 00:00:01

Time: 00:00:02

Time: 00:00:03

Enter your choice: 2

Stopwatch Stopped

Enter your choice: 3

Stopwatch Reset

Enter your choice: 4

Program Ended

FUTURE ENHANCEMENTS

1) Add graphical user interface (GUI) using Swing
2) Add lap timing feature
3) Store multiple time records
4) Add sound notifications
5) Convert into Android application
6) Add pause/resume functionality

GANTT CHART

Project Planning

Task	Duration
Requirement Analysis	2 Days
Design	3 Days
Coding	5 Days
Testing	3 Days
Documentation	2 Days

Time Management

- Week 1 → Planning & Design
- Week 2 → Coding
- Week 3 → Testing
- Week 4 → Documentation

CONCLUSION

- The Digital Stopwatch project successfully demonstrates the use of Java programming and Object-Oriented Programming concepts in developing a real-world application. The system provides accurate time measurement and is easy to use.
- This project enhances programming knowledge, improves logical thinking, and helps in understanding how real systems are developed. It also opens opportunities for further improvements and advanced implementations.

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